

Álvaro García Casas

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Education

Universidad Carlos III de Madrid (UC3M)

B.S. in Biomedical Engineering | GPA: 8.75 / 10.0

Expected May 2027

Georgia Institute of Technology (Academic Exchange)

B.S. in Biomedical Engineering | GPA: 4.00 / 4.00

May 2026

Stanford University School of Medicine

Continuing Medical Education (Online)

Sep 2025

Completed 300+ certified hours of advanced online training in medical innovation and clinical research, focusing on Artificial Intelligence in healthcare, diagnostic imaging technologies, and evidence-based clinical methodologies.

Experience

R&D Engineer, Georgia Tech Medical Robotics, Atlanta, GA

Aug 2025 – May 2026

- Contributed to the development of *Limbo*, a next-generation hand prosthesis integrating hardware, embedded acquisition, and real-time processing.
- Characterized and decoded residual neuromuscular activation patterns using Python and MATLAB pipelines for proportional myoelectric control.
- Optimized signal acquisition routines and digital filtering algorithms, achieving control accuracy exceeding 92% and improving clinical performance by 15%.

Co-Founder & Product Lead, Smart Stove Safety Knob (Create-X), Atlanta, GA

Jan 2026 – May 2026

- Spearheaded end-to-end development of an IoT safety device in Georgia Tech's venture accelerator, turning \$1,000 seed funding into a fully functional prototype preventing unattended-cooking fires.
- Engineered a low-power motion-based automatic shutoff mechanism with mobile app connectivity, optimizing unit manufacturing costs to ~\$15 for large-scale commercial rollout.

Technical Projects

Cardiac Electrophysiology and Peripheral Circulation Signal Analysis

Feb 2026 – May 2026

- Acquired synchronized single-lead ECG recordings and peripheral pulse waveforms using an IXTA data acquisition system, a ROAM ECG unit, and a PPG pulse plethysmograph.
- Identified and clinically interpreted P, QRS, and T wave components, correlating cardiac electrical events with peripheral mechanical circulation timing and physical exertion effects.
- Extended analysis to pulse oximetry (SpO₂), evaluating signal fidelity and reliability under thermoregulatory vasoconstriction and vasodilation states.

AI & LLM Web Platform for Clinical Report Extraction and Structuring

Jul 2026 – Aug 2026

- Developed a Python web application (LLMs, NLP) for automated ingestion, parsing, and structuring of complex pathology and biopsy reports (unstructured narrative and tables).
- Implemented in-memory, GDPR-compliant local anonymization pipelines and relational database/Excel synchronization modules, accelerating structured clinical metric extraction.

Wearable Sensor System for Stress and Tremor Analysis in Parkinson's Disease

Aug 2025 – Dec 2025

- Designed and prototyped a continuous-monitoring forearm cuff integrating surface EMG and GSR sensors to quantitatively assess physiological stress impacts on pathological tremor.
- Developed an Arduino acquisition and real-time MATLAB processing pipeline, validating 5 trials across 10 patients and detecting a statistically significant inverse correlation ($p = 0.041$, $R^2 = 0.4266$).

Skills & Certifications

- Technical Skills:** ECG, EMG, Signal Processing, Data Acquisition, Python, Pandas, PyTorch, Rapid Prototyping, SQL, Git/GitHub, AI APIs, Analog Electronics, Embedded Programming, MATLAB, CAD, Linear Regression, Mechatronics, Microsoft Office.
- Languages:** Spanish (Native), English (C1).
- Honors & Certifications:** Community of Madrid Excellence Award, Akademia Future Builders (Bankinter Innovation Foundation), MIT Regenerative AI, J&J Robotics, GT Design Expo Honorable Mention, Advanced Onshape Assemblies.

Community & Volunteering

Misión País Spain | Community Volunteer

Jan 2025 – Feb 2025

- Participated in a rural community mission alongside a multidisciplinary team of 30 volunteers, delivering support activities for individuals with disabilities, improving local accessibility, and assisting with on-site community initiatives.